A Bin-Packing Mean-Field Analysis

This appendix derives two expected-value results for the thresholded reinsertion policy of §4.2.1 under mean-field assumptions: the characteristic time between local repairs, and the steady-state active bin count.

A.1 Model and Assumptions

Let each model demand vector be $v_i \in [0,1]^3$, and let each bin have normalized capacity $(1,1,1)$. For model i, define its scalar size $s_i \triangleq \|v_i\|_1/3 \in [0,1]$ and write $v_i = s_i\mathbf{1} + \xi_i$ with $\langle \xi_i, \mathbf{1}\rangle = 0$. Then $\Phi(v_i) = \|\xi_i\|_2^2$. Let $\bar{s} \triangleq \mathbb{E}[s_i]$ and $\bar{\phi} \triangleq \mathbb{E}[\Phi(v_i)]$. For a bin B, let $r_B = \mathbf{1} - \sum_{i\in B} v_i$ be the residual vector, and define the damage ratio $\rho(B) \triangleq \Phi(r_B)/\sum_{i\in B}\Phi(v_i)$.

We use the following assumptions:

(A1) Model arrivals form a Poisson process with rate λ; each active model departs independently after an exponential lifetime with rate μ, giving $\mathbb{E}[N_{\text{act}}] = \lambda/\mu$.

(A2) The workload distribution is non-adversarial: $s_i \in [0,1]$, $\bar{s} \in (0,1)$, $\bar{\phi} \in (0,\infty)$.

(A3) Immediately after a local repack or global consolidation, a touched bin has expected damage ratio $\rho_{\text{reset}} \in (0,1)$.

(A4) The characteristic local threshold-crossing time is approximated by the first time at which the departure-only proxy trajectory reaches α.

(A5) At stationarity, if a random active bin B has residual r_B, then $\delta_B \triangleq \min_d r_{B,d}$ has approximately constant mean $\eta \triangleq \mathbb{E}[\delta_B] \in [0,1)$.

A.2 Repacking Frequency

Theorem A.1 (Characteristic repair time)**.** *Under Assumptions A1, A3, and A4, the characteristic local threshold-crossing time satisfies*

$$\tau_\alpha \;\approx\; \frac{1}{\mu}\log\!\left(\frac{1-\rho_{\text{reset}}}{1-\alpha}\right).$$

Proof. Fix a tagged bin immediately after a local repair, and let B_0 denote its resident set. Define $A_0 \triangleq \sum_{i\in B_0}\Phi(v_i)$ and $C_0 \triangleq \sum_{i\neq j\in B_0}\langle \xi_i,\xi_j\rangle$. Since $\Phi\big(\sum_{i\in B_0}\xi_i\big) = A_0 + C_0$, the initial damage ratio is $\rho_{\text{reset}} = 1 + C_0/A_0$. Because $\rho_{\text{reset}} < 1$, we have $C_0 < 0$, reflecting the complementarity created by repair.

Ignoring future arrivals and tracking only the surviving subset under Assumption A1, each model survives to time t with probability $q(t) = e^{-\mu t}$. The intrinsic and cross terms thin at different rates: $\mathbb{E}[A_t \mid B_0] = q(t)A_0$ and $\mathbb{E}[C_t \mid B_0] = q(t)^2 C_0$. The departure-only proxy damage ratio is therefore

$$\tilde{\rho}(t) = 1 - e^{-\mu t}(1-\rho_{\text{reset}}).$$

Setting $\tilde{\rho}(t) = \alpha$ and solving gives the claimed expression. □

Corollary A.2 (Effective reset interval)**.** *If the background global consolidation runs every $T_g = 30$ s, the effective reset interval is $\tau_\alpha^{\text{eff}} \approx \min\{\tau_\alpha, T_g\}$.*

Since departures arrive at rate μ, the amortized repacking cost per model is $1/(\mu\tau_\alpha) \approx 1/\log\!\big((1-\rho_{\text{reset}})/(1-\alpha)\big)$, an $O(1)$ constant confirming bounded migration churn.

A.3 Steady-State Bin Count

Theorem A.3 (Bin-count ratio)**.** *Under Assumptions A1, A2, and A5, together with the mean-field closure induced by the local threshold rule, the steady-state overhead ratio $\mathcal{R}_{\text{mf}} \triangleq \mathbb{E}[N]/U_0$ where $U_0 = \lambda\bar{s}/\mu$ is approximated by*

$$\mathcal{R}_{\text{mf}} \;\approx\; \left(\frac{\beta\sqrt{\zeta} + \sqrt{\beta^2\zeta + 4(1-\eta)}}{2(1-\eta)}\right)^2,$$

where $\beta \triangleq \sqrt{2\alpha/3}$ and $\zeta \triangleq \bar{\phi}/\bar{s}$.

Proof. Let $U = \sum_i s_i$ be the total normalized scalar volume and N the number of active bins. For each active bin B, define $\delta_B = \min_d r_{B,d}$ and $F_B = \|r_B\|_1 - 3\delta_B$. The scalar-capacity accounting identity is $N = U + \sum_B \delta_B + \frac{1}{3}\sum_B F_B$. Let $y_B = r_B - \delta_B\mathbf{1}$; one coordinate is zero and the other two are nonneg., giving $F_B = u_B + v_B$. Since Φ is shift-invariant along $\mathbf{1}$, $\Phi(r_B) = \Phi(y_B) \geq \frac{1}{6}(u_B+v_B)^2 = \frac{1}{6}F_B^2$, so $F_B \leq \sqrt{6\,\Phi(r_B)}$.

Define $A_B = \sum_{i\in B}\Phi(v_i)$ and $A = \sum_B A_B$. Under the local threshold rule, $\rho(B) \lesssim \alpha$, so $F_B \lesssim \sqrt{6\alpha\,A_B}$. Summing over bins and applying Cauchy–Schwarz: $\sum_B F_B \lesssim \sqrt{6\alpha\,N\,A}$.

Taking expectations in the accounting identity above with $\mathbb{E}[U] = U_0$ and $\mathbb{E}[A] = \lambda\bar{\phi}/\mu$, letting $x = \sqrt{\mathbb{E}[N]}$, and solving the resulting quadratic $(1-\eta)x^2 - \beta\sqrt{A_0}\,x - U_0 \approx 0$ gives the claimed formula after dividing by U_0 and substituting $\zeta = \bar{\phi}/\bar{s}$. □

In the light-fragmentation regime (small η and $\beta\sqrt{\zeta}$), this simplifies to $\mathcal{R}_{\text{mf}} \approx (1-\eta)^{-1} + \beta\sqrt{\zeta}\,(1-\eta)^{-3/2}$, confirming near-unit overhead when bins are well packed.

B LST-IMH Algorithm

B.1 Pseudocode

Algorithm 1 LST-IMH: Latest-Start-Time Iterative Moore–Hodgson

Require: Jobs J with (p_j, d_j); machines with availability times $T_1 \geq \cdots \geq T_m$.
Ensure: Feasible schedule and on-time job set S.

```
R ← J;   S ← ∅.
for i = 1, 2, ..., m do
    R_i ← {j ∈ R : d_j − T_i ≥ 0};   d_j^(i) ← d_j − T_i.
    S_i ← MOORE-HODGSON(R_i, {p_j}, {d_j^(i)}).
    Schedule S_i on M_i in EDD order from T_i.
    S ← S ∪ S_i;   R ← R \ S_i.
end for
return S.
```

Algorithm 2 Moore–Hodgson (single machine)

Require: Jobs R with processing times $\{p_j\}$ and deadlines $\{d_j\}$.
Ensure: Maximum-cardinality on-time job set.

```
Sort R by non-decreasing d_j (EDD order).
S ← ∅;   t ← 0;   H ← empty max-heap keyed on p_j.

for each j in EDD order do
    S ← S ∪ {j};   H.push(j);   t ← t + p_j.
    if t > d_j then
        j* ← H.pop() {largest p_j*}
        S ← S \ {j*};   t ← t − p_j*.
    end if
end for
return S.
```

B.2 Approximation Guarantee

Theorem B.1 (2-Approximation). *Let OPT be the maximum number of on-time jobs in an optimal schedule for $P_m \mid T_i \mid \sum U_j$, and let $ALG = |S|$ be the count returned by LST-IMH. Then $ALG \geq \frac{1}{2}\,OPT$.*

Proof. Fix an optimal schedule and let O_i denote the set of jobs completed on time on machine M_i in this optimal solution. The sets $O_1, \dots, O_m$ are pairwise disjoint and satisfy $\bigcup_{i=1}^{m} O_i = OPT$.

Consider iteration i of LST-IMH. The current residual job set is $R = J \setminus (S_1 \cup \cdots \cup S_{i-1})$. Among the jobs that the optimal solution assigns to M_i, those not yet claimed by earlier iterations form the subset

$$O_i \cap R \;=\; O_i \setminus (S_1 \cup \cdots \cup S_{i-1}).$$

Because the full set O_i is feasible on M_i starting at T_i (by optimality), removing some of its members can only relax the schedule: $O_i \cap R$ remains a feasible candidate for the single-machine sub-problem on M_i.

The algorithm invokes Moore–Hodgson on M_i, which returns a maximum-cardinality on-time subset for a single machine. Since $O_i \cap R$ is one feasible solution, we have

$$|S_i| \;\geq\; |O_i \cap R| \;=\; |O_i \setminus (S_1 \cup \cdots \cup S_{i-1})| \;\geq\; |O_i \setminus S|, \quad (1)$$

where the last inequality holds because $S \supseteq S_1 \cup \cdots \cup S_{i-1}$, so subtracting a larger set can only shrink the remainder.

Summing (1) over $i = 1, \dots, m$:

$$|S| = \sum_{i=1}^{m} |S_i| \;\geq\; \sum_{i=1}^{m} |O_i \setminus S|.$$

Because the O_i are pairwise disjoint, so are the $O_i \setminus S$, giving

$$\sum_{i=1}^{m} |O_i \setminus S| \;=\; \left|\bigcup_{i=1}^{m} (O_i \setminus S)\right| \;=\; \left|\left(\bigcup_{i=1}^{m} O_i\right) \setminus S\right| \;=\; |OPT \setminus S| \;=\; |OPT| - |OPT \cap S|.$$

Combining with the trivial bound $|OPT \cap S| \leq |S|$ yields

$$|S| \;\geq\; |OPT| - |S| \qquad \Longrightarrow \qquad 2|S| \;\geq\; |OPT|. \qquad \square$$

B.3 Remarks

Remark 1 (Order independence). The set-theoretic relaxation $|O_i \setminus (S_1 \cup \cdots \cup S_{i-1})| \geq |O_i \setminus S|$ holds for any processing order of the machines. Consequently, the $\frac{1}{2}$-approximation guarantee is valid regardless of how machines are sorted.

Remark 2 (Why LST ordering). Although the bound is order-independent, the LST (Latest Start Time) ordering is crucial in practice. A machine with a large availability time T_i has a restricted feasible domain: only jobs satisfying $d_j - T_i \geq p_j$ can be completed on time. If lightly loaded machines (small T_i) are processed first, they greedily consume "easy" jobs—those with generous slack that *any* machine could handle—while leaving tight-deadline jobs for the constrained machines. This can exhaust the constrained machines' feasible sets entirely, pushing performance to the worst-case $\frac{1}{2}\,OPT$ boundary. Processing the most constrained machine first reverses this dynamic: it reserves easy jobs for flexible machines that can absorb them later, consistently yielding near-optimal solutions in practice.